SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course: Artificial Intelligence | CO Assessed: CO2

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CO2 Statement: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently.

Note: Solutions below follow the question data and specified tie-breaking/search rules. Intermediate calculations and final answers are shown for each question.

QUESTION 1 – CAMPUS DELIVERY ROBOT

Topic: Informed Search Strategies and Heuristic Functions

Given: Road costs: S–A = 1, S–B = 4, A–C = 2, A–D = 5, B–G = 5, C–D = 1, D–G = 2. Heuristic values: h(S)=6, h(A)=5, h(B)=3, h(C)=3, h(D)=2, h(G)=0.

(a) Greedy Best-First Search

Greedy Best-First Search selects the frontier node with the smallest heuristic h(n), without considering the path cost already travelled.

Step	Expanded node	Frontier after expansion (ordered by h)
1	S	B (h=3), A (h=5)
2	B	G (h=0), A (h=5)
3	G – goal	Stop

Returned path: S → B → G
Total cost: S–B + B–G = 4 + 5 = **9 minutes**.

(b) A* Search

A* evaluates each node using f(n) = g(n) + h(n), where g(n) is the path cost from S and h(n) is the estimated remaining cost. The open list below is shown after each expansion; nodes are ordered by increasing f, with alphabetical tie-breaking.

Expanded	Open list after expansion: node (g,h,f)
S	A (1,5,6), B (4,3,7)
A	C (3,3,6), B (4,3,7), D (6,2,8)
C	D (4,2,6), B (4,3,7)
D	G (6,0,6), B (4,3,7)
G – goal	Stop; lowest-cost goal path is confirmed

When C is expanded, the route to D through C has g(D)=3+1=4, improving the earlier value g(D)=6 through A. D is updated/reopened with f(D)=4+2=6.

Expansion order: S → A → C → D → G
Final path: S → A → C → D → G

Total cost: 1 + 2 + 1 + 2 = **6 minutes**.

(c) Admissibility and consistency

The true shortest remaining costs to G are: G=0, D=2, C=3, A=5, B=5, S=6. A heuristic is admissible when h(n) never exceeds the true remaining cost h*(n).

Node	h(n)	True cost h*(n)	Admissible?
S	6	6	Yes
A	5	5	Yes
B	3	5	Yes
C	3	3	Yes
D	2	2	Yes
G	0	0	Yes

Consistency requires $h(u) \leq c(u,v) + h(v)$ for every directed road $u \rightarrow v$.

Road	Check	Result
S–A	$6 \leq 1+5 = 6$	Consistent
S–B	$6 \leq 4+3 = 7$	Consistent
A–C	$5 \leq 2+3 = 5$	Consistent
A–D	$5 \leq 5+2 = 7$	Consistent
B–G	$3 \leq 5+0 = 5$	Consistent
C–D	$3 \leq 1+2 = 3$	Consistent
D–G	$2 \leq 2+0 = 2$	Consistent

Conclusion: The heuristic is both admissible and consistent.

(d) Comparison and deployment

Greedy Best-First Search returns a path of cost 9 because it prioritizes only h(n); it chooses B because h(B)=3, even though the route through B is expensive. A* considers both the cost already paid and the estimated remaining cost, and finds the path of cost 6. For a delivery robot where travel cost matters, **A* is preferred** because this admissible and consistent heuristic supports an optimal path.

QUESTION 2 – WI-FI ACCESS POINT PLACEMENT

Topic: Local Search, Optimization and Online Search

Signal qualities: f(0)=2, f(1)=5, f(2)=9, f(3)=6, f(4)=4, f(5)=7, f(6)=12, f(7)=10, f(8)=8, f(9)=3. A move can only go to an adjacent position; hill climbing moves to a strictly better neighbour.

(a) Steepest-ascent hill climbing

Start	Step-by-step moves	Final position	Final quality
x=0	0 (2) → 1 (5) → 2 (9); stop because f(3)=6 is lower	2	9
x=4	4 (4) → 5 (7) → 6 (12); stop because f(7)=10 is lower	6	12
x=9	9 (3) → 8 (8) → 7 (10) → 6 (12); stop	6	12

(b) Local maxima, global maximum and restart probabilities

A local maximum is better than its immediate neighbours. Position x=2 has quality 9 and is a local maximum; position x=6 has quality 12 and is the global maximum. The remaining positions are not local maxima.

Quantity	Calculation	Answer
Local maxima	x=2 (quality 9); x=6 (quality 12)	x=2, x=6
Probability one run reaches global maximum	Starts x=4,5,6,7,8,9 reach x=6: 6/10	0.6 = 60%
At least one of 3 independent random-restart runs	1 – P(all 3 fail) = 1 – (1–0.6)^3 = 1–0.064	0.936 = 93.6%
Expected number of runs	Geometric expectation = 1/p = 1/0.6	5/3 ≈ 1.67 runs

(c) Simulated annealing acceptance probability

The move is from x=2, where f=9, to x=3, where f=6. Thus Δ = 6–9 = –3. For a worse move, simulated annealing accepts with probability e^(Δ/T).

Temperature T	Calculation	Acceptance probability
2	e^(–3/2)	≈ 0.2231 (22.31%)
1	e^(–3)	≈ 0.0498 (4.98%)

As temperature falls, the probability of accepting this worse move decreases. The search becomes less willing to move downhill, so it becomes harder to escape a local maximum late in the search.

(d) Why this is online search

It is an online search problem because the robot does not know the full corridor or all signal values in advance. It observes f(x) only at its current location and must choose actions while discovering the environment; exploration also has a cost. To avoid cycling, the agent should remember visited positions, previous states/actions and relevant measured signal values (or maintain a map of explored positions).

QUESTION 3 – EXAM TIMETABLE SCHEDULING

Topic: Constraint Satisfaction Problem (CSP)

Variables: A, B, C, D, E. Domains: A={1,2}, B={1,2,3}, C={1,2,3}, D={1,2,3}, E={1,2}. Adjacent exams in the constraint graph must have different time slots.

(a) Constraint graph and degrees

Edges: A–B, A–C, A–D, B–C, B–D, C–E, D–E.

Exam	Neighbours	Degree
A	B, C, D	3
B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

(b) First variable and assignment

MRV selects a variable with the smallest domain. A and E both have 2 values, so MRV ties. The degree heuristic chooses A because it has 3 neighbours, while E has 2. Alphabetical tie-breaking is not needed. Assign A=1 (the lowest value). Forward checking removes slot 1 from A's unassigned neighbours B, C and D.

After assignment	A	B	C	D	E
A=1	{1}	{2,3}	{2,3}	{2,3}	{1,2}

(c) Continue search using MRV, degree heuristic, alphabetical tie-break and ascending values

Step	Choice and value	Domains after forward checking
1	A=1	A={1}; B={2,3}; C={2,3}; D={2,3}; E={1,2}
2	B=2	A={1}; B={2}; C={3}; D={3}; E={1,2}
3	C=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
4	D=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
5	E=1	Complete assignment found

Explanation of selection order: After A=1, B, C and D each have 2 remaining values and each has 2 unassigned neighbours; alphabetical order selects B. Assigning B=2 removes 2 from C and D, leaving each with {3}. MRV then selects C alphabetically, followed by D. E is the final variable and its lowest available value is 1.

Final timetable: A=1, B=2, C=3, D=3, E=1.

Constraint verification: A differs from B, C and D (1 differs from 2, 3 and 3); B differs from C and D (2 differs from 3); C differs from E (3 differs from 1); and D differs from E (3 differs from 1). Therefore, every stated constraint is satisfied.

(d) Interpretation

MRV tries the most constrained variables first, reducing the chance of discovering a conflict late. The degree heuristic breaks MRV ties by selecting the variable constraining the largest number of other unassigned variables. Forward checking removes values that conflict with each assignment and exposes conflicts early. If any unassigned variable's domain becomes empty, the current branch cannot lead to a solution, so the algorithm must backtrack.

QUESTION 4 – BOARD-GAME AI

Topic: Minimax, Alpha-Beta Pruning and Evaluation Functions

Evaluation function: Eval = 2 × M + Mo, where M is MAX's material advantage and Mo is MAX's mobility advantage. MAX chooses among B, C and D; MIN then chooses the minimum-valued leaf under each move.

(a) Evaluation of all nine leaves

MIN node	Leaf 1 (M,Mo)	Eval	Leaf 2 (M,Mo)	Eval	Leaf 3 (M,Mo)	Eval
B	(1,1)	2×1+1=3	(5,2)	2×5+2=12	(3,2)	2×3+2=8
C	(1,0)	2×1+0=2	(1,2)	2×1+2=4	(2,2)	2×2+2=6
D	(6,2)	2×6+2=14	(2,1)	2×2+1=5	(0,2)	2×0+2=2

(b) Plain minimax

MIN chooses the smallest leaf value below each move. MAX then chooses the largest of those backed-up values.

MAX move	Leaf values	MIN backed-up value
B	3, 12, 8	min = 3
C	2, 4, 6	min = 2
D	14, 5, 2	min = 2

Root value: max(3, 2, 2) = **3**. **Best move for MAX:** B.

(c) Alpha-beta pruning, searching left to right

Start at the root MAX node with $\alpha = -\infty$ and $\beta = +\infty$. After evaluating B, its MIN value is 3, so the root's α becomes 3.

Node	Leaf evaluations / bounds	Pruning decision
B (MIN)	Evaluate 3, then 12, then 8; β updates $3 \to 3 \to 3$. B value = 3.	No pruning in B.
Root MAX	$\alpha = \max(-\infty, 3) = 3$.	Continue to C.
C (MIN)	First leaf = 2; $\beta = 2$. Since $\beta \le \alpha$ ($2 \le 3$)...	Prune C's remaining leaves 4 and 6.
D (MIN)	Evaluate 14 (β=14), then 5 (β=5), then 2 (β=2).	No leaves remain to prune under D.

Leaves evaluated: B uses 3 leaves, C uses 1 leaf, and D uses 3 leaves, for a total of **7 leaves**. The root value remains **3**, and MAX chooses **B**.

(d) If C's leaves are ordered (6, 4, 2)

With root α=3, C first evaluates 6 (β=6, no cutoff because 6>3), then 4 (β=4, still no cutoff because 4>3), then 2 (β=2, cutoff condition $\beta \le \alpha$ is met). All three C leaves must be evaluated. B and D also require three evaluations each in this left-to-right tree, so alpha-beta evaluates **9 leaves**.

Move ordering affects how early a cutoff can be found: a low value encountered early at a MIN node can trigger pruning sooner when α is already high. Alpha-beta pruning never changes the minimax value because it only skips branches that cannot affect the final choice under the current α and β bounds.

END OF ANSWERS

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